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Wireless communication network optimization for user applications in wireless user equipment (UE)

Abstract

A wireless communication device optimizes network communications for a user application. The wireless communication device generates network characteristics for multiple wireless communication networks based on wireless signals received from the multiple wireless communication networks. The wireless communication device transfers the network characteristics for the multiple wireless communication networks to the user application. The wireless communication device receives a selection of one of the wireless communication networks from the user application based on the network characteristics. The wireless communication device communicates over a radio with the selected one of the multiple wireless communication networks responsive to the selection from the user application.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8964595	12/2014	Agarwal et al.	N/A	N/A
9693366	12/2016	Adjakple et al.	N/A	N/A
9924386	12/2017	Jain et al.	N/A	N/A
10212579	12/2018	Song et al.	N/A	N/A
10686871	12/2019	Taaghol et al.	N/A	N/A
2003/0100308	12/2002	Rusch	455/445	H04W 48/18
2006/0015636	12/2005	Skraba et al.	N/A	N/A
2010/0162348	12/2009	Narayanan et al.	N/A	N/A
2013/0109370	12/2012	Rowson et al.	N/A	N/A
2017/0150470	12/2016	Kalapatapu et al.	N/A	N/A
2017/0245316	12/2016	Salkintzis	N/A	N/A
2017/0302522	12/2016	Tonshal et al.	N/A	N/A
2018/0359689	12/2017	Lee et al.	N/A	N/A
2019/0053139	12/2018	Basu Mallick et al.	N/A	N/A
2019/0373301	12/2018	Gunasekara et al.	N/A	N/A
2020/0059856	12/2019	Cui	N/A	H04W 24/02
2020/0245160	12/2019	Chu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3402254	12/2019	EP	N/A
2020091636	12/2019	WO	N/A

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Background/Summary

RELATED CASES (1) This United States Patent Application is a continuation of U.S. patent application Ser. No. 17/166,127 that was filed on Feb. 3, 2021 and is entitled “WIRELESS COMMUNICATION NETWORK OPTIMIZATION FOR USER APPLICATIONS IN WIRELESS USER EQUIPMENT (UE).” U.S. patent application Ser. No. 17/166,127 is hereby incorporated by reference into this United States Patent Application.

TECHNICAL BACKGROUND

(1) Wireless communication networks provide wireless data services to wireless user devices. Exemplary wireless data services include machine-control, internet-access, media-streaming, and social-networking. Exemplary wireless user devices comprise phones, computers, vehicles, robots, and sensors. The wireless communication networks have wireless access nodes which exchange wireless signals with the wireless user devices over radio frequency bands. The wireless signals use wireless network protocols like Fifth Generation New Radio (5G NR), Long Term Evolution (LTE), Institute of Electrical and Electronic Engineers (IEEE) 802.11 (WIFI), and Low-Power Wide Area Network (LP-WAN). The wireless access nodes exchange network signaling and user data with network elements that are often clustered together into wireless network cores. The wireless access nodes are connected to the wireless network cores over backhaul data links.

(2) The wireless access nodes comprise Radio Units (RUs), Distributed Units (DUs) and Centralized Units (CUs). The RUs are mounted at elevation and have antennas, modulators, signal processor, and the like. The RUs are connected to the DUs which are usually nearby network computers. The DUs handle lower wireless network layers like the Physical Layer (PHY) and Media Access Control (MAC). The DUs are connected to the CUs which are larger computer centers that are closer to the network cores. The CUs handle higher wireless network layers like the Radio Resource Control (RRC) and Packet Data Convergence Protocol (PDCP). The CUs are coupled to network elements in the network cores. Exemplary network elements include Access and Mobility Management Functions (AMFs), Session Management Functions (SMFs), and User Plane Functions (UPFs).

(3) The wireless user devices comprise user circuitry and wireless network circuitry. The wireless network circuitry has radios and network applications to communicate over the air with the RUs in the wireless access nodes. The user circuitry has an operating system and user applications that use the wireless network circuitry for external data communications. The wireless network circuitry receives and processes wireless signals from the wireless access nodes to develop network characteristics for wireless communication networks. The network characteristics describe wireless network types, wireless signal qualities, Public Land Mobile Network (PLMN) features, wireless network slice parameters, UE antenna configurations, Radio Resource Control (RRC) status, System Information Block (SIB) broadcasts, and the like.

(4) To optimize network communications, a user application in a wireless user device calls its operating system for information that indicates wireless network types and their wireless signal qualities. In response to the call, the operating system transfers a request for the wireless network types and their wireless signal qualities to the wireless network circuitry. The wireless network circuitry responds to the operating system with the wireless network types and their wireless signal qualities. The operating system transfers the wireless network types and their wireless signal qualities to the user application. The user application selects a wireless communication network based on the wireless network types and their wireless signal qualities. The user application then transfers a call to the operating system for the selected wireless communication network. The operating system transfers a request for the selected wireless communication network to the wireless network circuitry. The wireless network circuitry exchanges user data for the user

application with the selected wireless communication network.

(5) Unfortunately, the user application does not have access to detailed network characteristics like PLMN features and network slice parameters when selecting its wireless communication networks. Moreover, the user application does not optimize its network communications based on detailed network characteristics like PLMN features and network slice parameters.

TECHNICAL OVERVIEW

(6) In some examples, network communications are optimized for a user application. Network characteristics are generated for multiple wireless communication networks based wireless signals received from the multiple wireless communication networks. The network characteristics for the multiple wireless communication networks are transferred to the user application. A selection of one of the multiple wireless communication networks is received from the user application based on the network characteristics. The selected wireless communication network is communicated with over a radio responsive to the selection from the user application.

(7) In some examples, one or more non-transitory machine-readable storage media stores processing instructions that direct processing circuitry to perform a method to optimize network communications for a user application. The method comprises generating network characteristics for multiple wireless communication networks based wireless signals received from the multiple wireless communication networks. The method comprises transferring the network characteristics for the multiple wireless communication networks to the user application. The method comprises receiving a selection of one of the multiple wireless communication networks from the user application based on the network characteristics. The method comprises communicating over a radio with the selected one of the multiple wireless communication networks responsive to the selection from the user application.

(8) In some examples, a wireless communication device optimizes network communications for a user application. The wireless communication device comprises network circuitry and processing circuitry. The network circuitry wirelessly receives wireless signals from multiple wireless communication networks, and in response, generates network characteristics for the multiple wireless communication networks and transfers the network characteristics for the multiple wireless communication networks to the processing circuitry. The processing circuitry transfers the network characteristics to the user application and responsively receives a selection of one of the one of the multiple wireless communication networks from the user application. The processing circuitry exchanges user data with the user application and the network circuitry. The network circuitry wirelessly exchanges the user data with the selected one of the multiple wireless communication networks.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a wireless User Equipment (UE) to optimize network communications for user applications.

(2) FIG. 2 illustrates the operation of wireless UE to optimize network communications for the user applications.

(3) FIG. 3 further illustrates the operation of wireless UE to optimize network communications for the user applications.

(4) FIG. 4 illustrates a Fifth Generation (5G) UE to optimize network communications for user applications.

(5) FIG. 5 illustrates the 5G UE which optimizes network communications for the user applications.

(6) FIG. 6 illustrates a 5G Radio Access Network (RAN) that serves the 5G UE which optimizes

network communications for the user applications.

(7) FIG. 7 illustrates a 5G wireless network core that serves the 5G UE which optimizes network communications for the user applications.

(8) FIG. 8 further illustrates the 5G wireless network core that serves the 5G UE which optimizes network communications for the user applications.

(9) FIG. 9 illustrates the operation of the 5G UE to optimize network communications for the user applications.

DETAILED DESCRIPTION

(10) FIG. 1 illustrates wireless User Equipment (UE) **101** to optimize network communications for user application **105**. Wireless UE **101** comprises a computer, phone, vehicle, sensor, robot, or some other data appliance with wireless communication circuitry. Wireless UE **101** consumes wireless data services like internet-access, machine communications, media-conferencing, or some other wireless data product. Wireless UE **101** comprises wireless network circuitry **102** and user circuitry **103**. User circuitry **103** comprises operating system **104** and user application **105**. Other user applications are typically present and operate like user application **105** but the other user applications are omitted for clarity.

(11) Various examples of UE operation and configuration are described herein. In some examples, operating system **104** receives a user permission for user application **105** to access network characteristics for wireless communication networks **111-113**. For example, operating system **104** could direct a graphic display to prompt a human user to permit user application **105** to access network characteristics for networks **111-113** from wireless network circuitry **102**. Exemplary network characteristics describe: Public Land Mobile Networks (PLMNs), wireless signal quality, wireless network slices, UE antenna configurations, Radio Resource Control (RRC) status, System Information Block (SIB) broadcasts, and the like. To optimize network communications, user application **105** transfers a call to operating system **104** for the network characteristics. In response to the user permission and the call, operating system **104** transfers a request for the network characteristics to wireless network circuitry **102**. Wireless network circuitry **102** receives and processes wireless signals from wireless communication networks **111-113** to generate individual network characteristics for wireless communication networks **111-113**. Wireless network circuitry **102** responds to operating system **104** with the individual network characteristics. Operating system **104** transfers the network characteristics to user application **105**. User application **105** selects one or more of wireless communication networks **111-113** based on the network characteristics. User application **105** may also select Public Land Mobile Networks (PLMNs) and/or wireless network slices based on the network characteristics. User application **105** transfers a call to operating system **104** for the selected wireless communication networks **111-113**, PLMNs, and/or slices. Operating system **104** transfers a request for the selected wireless communication networks **111-113**, PLMNs, and/or slices to wireless network circuitry **102**. Wireless network circuitry **102** exchanges user data for user application **105** with user circuitry **103**. Wireless network circuitry **102** wirelessly exchanges user data for user application **105** with the selected wireless communication networks **111-113** using the selected PLMNs and/or slices.

(12) Advantageously, user application **105** uses detailed network characteristics like PLMN features and network slice parameters to select one or more wireless communication networks. Moreover, user application **105** optimizes its network communications based on the detailed network characteristics like PLMN features and network slice parameters.

(13) UE **101** and RANs **111-113** wirelessly communicate over wireless links using Radio Access Technologies (RATs) like Fifth Generation New Radio (5G NR), Long Term Evolution (LTE), Institute of Electrical and Electronic Engineers (IEEE) 802.11 (WIFI), Low-Power Wide Area Network (LP-WAN), and/or some other wireless protocol. The RATs use electromagnetic frequencies in the low-band, mid-band, high-band, or some other portion of the electromagnetic spectrum. RANs **111-113** communicate with network cores over backhaul links **121-123** that use

metallic links, glass fibers, radio channels, or some other communication media. Links **121-123** use IEEE 802.3 (Ethernet), Time Division Multiplex (TDM), Data Over Cable System Interface Specification (DOCSIS), Internet Protocol (IP), General Packet Radio Service Transfer Protocol (GTP), 5G NR, LTE, WIFI, Fifth Generation Core (5GC), virtual switching, inter-processor communication, bus interfaces, and/or some other data communication protocols. RANs **111-113** are depicted as towers but RANs **111-113** may use other mounting structures or no mounting structures at all. RANs **111-113** may comprise gNodeBs, eNodeBs, NB-IoT access nodes, LP-WAN base stations, wireless relays, and/or some other wireless network transceivers. UE **101** comprises antennas, amplifiers, filters, modulation, analog/digital interfaces, microprocessors, software, memories, transceivers, bus circuitry, and the like. The microprocessors comprise Digital Signal Processors (DSP), Central Processing Units (CPU), Graphical Processing Units (GPU), Application-Specific Integrated Circuits (ASIC), and/or the like. The memories comprise Random Access Memory (RAM), flash circuitry, disk drives, and/or the like. The memories store software like operating systems, user applications, radio applications, and network functions. The microprocessors retrieve the software from the memories and execute the software to drive the operation of wireless UE **101** and RANs **111-113** as described herein.

(14) FIG. 2 illustrates the operation wireless UE **101** to optimize network communications for user application **105**. This operation is exemplary and may vary in other examples. Operating system **104** receives a user permission for user application **105** to access network characteristics for wireless communication networks **111-113** (**201**). To optimize network communications, user application **105** transfers a call to operating system **104** for the network characteristics (**202**). In response to the user permission and the call (**203**), operating system **104** transfers a request for the network characteristics to wireless network circuitry **102** (**204**). Operating system **104** denies the call for the network characteristics from user application **105** when the user permission for from user application **105** is not present (**210**).

(15) Wireless network circuitry **102** receives and processes wireless signals from wireless communication networks **111-113** to generate and transfer network characteristics for wireless communication networks **111-113** to operating system **104**. Operating system **104** transfers the network characteristics to user application **105** (**206**). User application **105** selects one or more of wireless communication networks **111-113**, PLMNs, and/or wireless network slices based on the network characteristics (**207**). User application **105** transfers a call to operating system **104** for the selected wireless communication networks **111-113**, PLMNs, and/or slices (**207**). Operating system **104** transfers a request for the selected wireless communication networks **111-113**, PLMNs, and/or slices to wireless network circuitry **102** (**208**). Wireless network circuitry **102** exchanges user data for user application **105** with user circuitry **103** (**209**). Wireless network circuitry **102** wirelessly exchanges user data for user application **105** with the selected wireless communication networks **111-113**, PLMNs, and/or slices (**209**).

(16) FIG. 3 further illustrates the operation wireless UE **101** to optimize network communications for user application **105**. This operation is exemplary and may vary in other examples. Operating system **104** receives a user permission for user application **105** to access network characteristics. Exemplary network characteristics comprise information for wireless signal quality, PLMNs, wireless network slices, UE antenna configurations, RRC information, SIB data, and the like. To optimize network communications, user application **105** transfers a call to operating system **104** for the network characteristics. Operating system **104** verifies the user permission for user application **104** to access the network characteristics. In response to the user permission and the call, operating system **104** transfers a request for the network characteristics to wireless network circuitry **102**.

(17) Wireless network circuitry **102** receives and processes wireless signals from wireless communication networks **111-113** to generate individual network characteristics for wireless communication networks **111-113**. Wireless network circuitry **102** responds to operating system **104** with the individual network characteristics. Operating system **104** transfers the network

characteristics to user application **105**. In this example, user application **105** selects wireless communication network **112**, representative PLMN “A”, and representative wireless slice “B” based on the network characteristics. Other selections could be made in other examples. User application **105** transfers a call to operating system **104** for selected wireless communication network **112**, PLMN A, and slice B. Operating system **104** transfers a request for wireless communication network **112**, PLMN A, and slice B to wireless network circuitry **102**. Wireless network circuitry **102** exchanges user data for user application **105** with user circuitry **103**. Wireless network circuitry **102** wirelessly exchanges user data for user application **105** with wireless communication network **112** using PLMN A and slice B.

(18) FIG. 4 illustrates Fifth Generation (5G) UE **401** to optimize network communications for user application **405**. 5G UE **401** comprises an example of UE **101**, although UE **101** may differ. UE **401** comprises processing circuitry **403** and radios **407-409**. Processing circuitry **403** comprises operating system (OS) **404**, user application (USER) **405**, and network applications (NET) **406**. Radios **407-409** are wirelessly coupled to respective Radio Access Networks (RANs) **411-413** over wireless communication links. RANs **411-413** are coupled to respective wireless network cores **421-423** over backhaul communication links. RANs **411-413** comprise respective Radio Units (RUS) **431-433**, Distributed Units (DUs) **441-443**, and Centralized Units **451-453**.

(19) In wireless UE **401**, operating system **404** directs a graphic display to prompt the user and obtain the user's express permission for user application **405** to access network characteristics for RANs **411-413** and network cores **421-423**. Operating system **104** receives a user instruction through the display to permit user application **405** to access the network characteristics. The network characteristics comprise radio signal metrics, Public Land Mobile Network (PLMN) features, wireless network slice parameters, UE antenna configurations, Radio Resource Control (RRC) information, System Information Block (SIB) data, and the like. For example, the SIB data may indicate available PLMNs and their network slices along with the features of the PLMNs and slices.

(20) To optimize network communications, user application **405** transfers a call to operating system **404** for the network characteristics. In response to the user permission and the call, operating system **404** transfers a request for the network characteristics to network applications **406**. Network applications **406** transfer corresponding requests for the network characteristics to radios **407-409**. Radios **407-409** receive and process wireless signals from RANs **411-413** to generate wireless network symbols for RANs **411-413**. Radios **407-409** transfer the wireless network symbols for RANs **411-413** to network applications **406**. Network applications **406** process the network symbols to recover user data and network signaling. Network applications **406** process the network signaling to generate individual network characteristics for RANs **411-413**. Network applications **406** respond to operating system **404** with the individual network characteristics. Operating system **404** transfers the individual network characteristics to user application **405**.

(21) User application **405** selects RANs, PLMNs, and/or slices based on the network characteristics. For example, user application **405** may comprise an augmented reality application that selects a specific wireless network slice in a particular PLMN in RAN **413** because that slice has the best latency and mobility for the augmented reality service. In another example, user application **405** may comprise a robot communication application that selects a different wireless network slice in another PLMN in RAN **412** because that slice has the best range and throughput for the robot control service.

(22) In this example, user application **405** selects RAN **411**, PLMN A, and wireless network slice B. User application **405** transfers a call to operating system **404** for RAN **411**, PLMN A, and slice B. Operating system **404** transfers a request for RAN **411**, PLMN A, and slice B to one of network applications **406** that interacts with RAN **411**. The network application uses radio **407** to establish wireless data service for UE **401** and user application **405** over RAN **411** using PLMN A and slice B. User application **405** then exchanges user data over operating system **404**, network applications

406, radio **407**, RAN **411**, and wireless network core **421**. RAN **411** and wireless network core **421** use PLMN A and wireless network slice B to serve user application **405** in UE **401**. (23) FIG. 5 illustrates 5G UE **401** which optimizes network communications for user application **405**. UE **401** comprises an example of UE **101**, although UE **101** may differ. UE **401** comprises 5G NR radios **407-409** and processing circuitry **403**. 5G NR radios **407-409** comprise antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, and transceivers (XCVRs) that are coupled over bus circuitry. Processing circuitry **403** comprises memory, CPU, user interfaces and components, and transceivers that are coupled over bus circuitry. The memory in processing circuitry **403** stores operating system **404**, user applications (USER) **404**, and network applications (NET) **406**. Network applications comprise 5G NR Physical Layer (PHY), Media Access Control (MAC), Radio Link Control (RLC), Packet Data Convergence Protocol (PDCP), Service Data Adaptation Protocol (SDAP), and Radio Resource Control (RRC). The antennas in 5G NR radio **407-409** are wirelessly coupled to respective 5G NR RANs **411-413** over carrier frequency bands. Transceivers in 5G NR radios **407-409** are coupled to transceivers in processing circuitry **403**. Transceivers in user circuitry **403** are typically coupled to user components like displays, controllers, and memory. The CPU in processing circuitry **403** executes operating system **404**, user applications **405**, and network applications **406** to exchange 5G NR signaling and data with 5G NR RANs **411-413** over 5G NR radios **407-409**. To correlate UE **401** with UE **101**, radios **407-409** and portions of processing circuitry **403** that support network applications **406** represent wireless network circuitry **102**, while portions of processing circuitry **403** that support operating system **404** and user applications **405** represent user circuitry **103** in UE **101**.

(24) In 5G NR radios **407-409**, the antennas receive wireless signals from RANs **411-413** that transport downlink 5G NR signaling and data. The antennas transfer corresponding electrical signals through duplexers to the amplifiers. The amplifiers boost the received signals for filters which attenuate unwanted energy. Demodulators down-convert the amplified signals from their carrier frequency. The analog/digital interfaces convert the demodulated analog signals into digital signals for the DSPs. The DSPs transfer corresponding 5G NR symbols to processing circuitry **403** over the transceivers. In processing circuitry **403**, the CPU executes network applications **406** to process the 5G NR symbols and recover the downlink 5G NR signaling and data. Network applications **406** process the downlink 5G NR signaling to identify individual network characteristics for RANs **411-413**. Network applications **406** transfer the downlink user data to user applications **405** over operating system **404**.

(25) Network applications **406** receive uplink user data from user applications **405** over operating system **404**. Network applications **406** generate new uplink 5G NR signaling for the uplink user data. Network applications **406** process the uplink 5G NR signaling and user data to generate corresponding uplink 5G NR symbols that carry the uplink 5G NR signaling and user data. In 5G NR radios **407-409**, the DSP process the uplink 5G NR symbols to generate corresponding digital signals for the analog-to-digital interfaces. The analog-to-digital interfaces convert the digital uplink signals into analog uplink signals for modulation. Modulation up-converts the uplink analog signals to their carrier frequencies. The amplifiers boost the modulated uplink signals for the filters which attenuate unwanted out-of-band energy. The filters transfer the filtered uplink signals through duplexers to the antennas. The electrical uplink signals drive the antennas to emit corresponding wireless 5G NR signals to 5G RANs **411-413** that transport the uplink 5G NR signaling and user data.

(26) RLC functions comprise Automatic Repeat Request (ARQ), sequence numbering and resequencing, segmentation and resegmentation. MAC functions comprise buffer status, power control, channel quality, Hybrid Automatic Repeat Request (HARQ), user identification, random access, user scheduling, and QoS. PHY functions comprise packet formation/deformation, guard-insertion/guard-deletion, parsing/de-parsing, control insertion/removal, interleaving/de-

interleaving, Forward Error Correction (FEC) encoding/decoding, channel coding/decoding, channel estimation/equalization, and rate matching/de-matching, scrambling/descrambling, modulation mapping/de-mapping, layer mapping/de-mapping, precoding, Resource Element (RE) mapping/de-mapping, Fast Fourier Transforms (FFTs)/Inverse FFTs (IFFTs), and Discrete Fourier Transforms (DFTs)/Inverse DFTs (IDFTs). PDCP functions include security ciphering, header compression and decompression, sequence numbering and re-sequencing, de-duplication. SDAP functions include QoS marking and flow control. RRC functions include authentication, security, handover control, status reporting, QoS, network broadcasts and pages, and network selection.

(27) Operating system **404** directs a graphic display in the user components to prompt a user and obtain the user's express permission for user application **405** to access network characteristics for RANs **411-413**. Operating system **404** receives a user instruction through the display to permit user application **405** to access the network characteristics. The network characteristics comprise radio signal metrics, PLMN features, wireless network slice parameters, UE antenna configurations, RRC information, SIB data, and the like.

(28) To optimize network communications, user application **405** transfers a call to operating system **404** for the network characteristics. In response to the user permission and the call, operating system **404** transfers a request for the network characteristics to the RRCs in network applications **406** for radios **407-409**. The RRCs in network applications **406** receive 5G NR symbols over respective radios **407-409** from respective RANs **411-413**. The RRCs in network applications **406** process the symbols to derive the network characteristics for respective RANs **411-413**. The RRCs in network applications **406** respond to operating system **404** with the requested network characteristics. Operating system **404** transfers the requested network characteristics to user application **405**.

(29) User application **405** selects one or more RANs, PLMNs, and/or wireless network slices based on the network characteristics. In this example, user application **405** selects RAN **411**, PLMN A, and wireless network slice B. User application **405** transfers a call to operating system **404** for RAN **411**, PLMN A, and slice B. Operating system **404** transfers a request for RAN **411**, PLMN A, and slice B to the RRC in network applications **406** that interacts with RAN **411** over radio **407**. The RRC in network applications **406** uses radio **407** to establish wireless data service for UE **401** and user application **405** over RAN **411** using PLMN A and slice B. User application **405** then exchanges user data over operating system **404**, network application **406**, radio **407**, and RAN **411**. RAN **411** uses PLMN A and wireless network slice B to serve user application **405** in UE **401**.

(30) FIG. 6 illustrates 5G Radio Access Network (RAN) **411** that serves 5G UE **410** which optimizes network communications for user application **405**. RAN **411** comprises an example of RANs **111-113**, although RANs **111-113** may differ. RANs **412** and **413** would be similar to RAN **411**. RAN **411** comprises 5G Radio Unit (RU) **431**, 5G Distributed Unit (DU) **441**, and 5G Centralized Unit (CU) **451**. RU **431** comprises antennas, amplifiers, filters, modulation, analog-to-digital interfaces, DSP, memory, and transceivers (XCVRs) that are coupled over bus circuitry. DU **441** comprises memory, CPU, and transceivers that are coupled over bus circuitry. The memory in DU **441** stores operating systems and 5G NR network applications like Physical Layer (PHY), Media Access Control (MAC), and Radio Link Control (RLC). CU **451** comprises memory, CPU, and transceivers that are coupled over bus circuitry. The memory in CU **451** stores an operating system and network applications like Packet Data Convergence Protocol (PDCP), Service Data Adaptation Protocol (SDAP), and Radio Resource Control (RRC).

(31) UE **401** is wirelessly coupled to the antennas in RU **431** over 5G NR links. Transceivers in RU **431** are coupled to transceivers in DU **441** over fronthaul links like enhanced Common Public Radio Interface (eCPRI). Transceivers in DU **441** are coupled to transceivers in CU **451** over mid-haul links. Transceivers in CU **451** are coupled to wireless network core **421** over backhaul links. The DSP in RU **431** executes an operating system and radio applications to exchange 5G NR signals with UE **401** and to exchange 5G NR data units with DU **441**. The CPU in DU **441** executes

an operating system and PHY, MAC, and RLC to exchange 5G NR data units with RU **431** and to exchange 5G NR data units with CU **451**. The CPU in CU **451** executes an operating system and PDCP, SDAP, and RRC to exchange 5G signaling and data with wireless network core **421**.

(32) On the uplink, the antennas in RU **431** receive wireless signals from UE **401** that transport uplink 5G NR signaling and data. The antennas transfer corresponding electrical signals through duplexers to the amplifiers. The amplifiers boost the received signals for filters which attenuate unwanted energy. Demodulators down-convert the amplified signals from their carrier frequency. The analog/digital interfaces convert the demodulated analog signals into digital signals for the DSP. The DSP transfers corresponding 5G NR symbols to DU **441** over the transceivers. The PHY, MAC, and RLC in DU **441** process the uplink 5G NR symbols to generate Service Data Units (SDUs) which are transferred to the PDCP in CU **451**. In CU **451**, the PDCP, SDAP, and RRC process the uplink SDUs and the downlink 5G signaling and data to generate uplink 5G signaling and data and downlink SDUs. The RRC in CU **451** exchanges N2/N1 signaling with network core **421**. The SDAP in CU **451** exchanges N3 data network core **421**.

(33) On the downlink, the SDAP in CU **451** transfer user data to the PDCP. The PDCP in CU **451** transfers downlink SDUs to the RLC in DU **441**. The RLC, MAC, and PHY process the downlink SDUs to generate downlink 5G NR symbols. The PHY in DU **441** transfers the downlink 5G NR symbols to the DSP in RU **431**. The DSP receives downlink 5G NR symbols from DU **441**. The DSP processes the downlink 5G NR symbols to generate corresponding digital signals for the analog-to-digital interfaces. The analog-to-digital interfaces convert the digital signals into analog signals for modulation. Modulation up-converts the analog signals to their carrier frequency. The amplifiers boost the modulated signals for the filters which attenuate unwanted out-of-band energy. The filters transfer the filtered electrical signals through duplexers to the antennas. The filtered electrical signals drive the antennas to emit corresponding wireless signals to 5G NR UE **401** that transport the downlink 5G NR signaling and user data.

(34) RRC functions comprise authentication, security, handover control, status reporting, QoS, network broadcasts and pages, and network selection. SDAP functions comprise QoS marking and flow control. PDCP functions comprise security ciphering, header compression and decompression, sequence numbering and re-sequencing, de-duplication. RLC functions comprise ARQ, sequence numbering and resequencing, segmentation and resegmentation. MAC functions comprise buffer status, power control, channel quality, HARQ, user identification, random access, user scheduling, and QoS. PHY functions comprise packet formation/deformation, windowing/de-windowing, guard-insertion/guard-deletion, parsing/de-parsing, control insertion/removal, interleaving/de-interleaving, FEC encoding/decoding, channel coding/decoding, channel estimation/equalization, and rate matching/de-matching, scrambling/descrambling, modulation mapping/de-mapping, layer mapping/de-mapping, precoding, RE mapping/de-mapping, FFTs/IFFTs, and DFTs/IDFTs.

(35) The RRC in CU **451** receives SIB data from network core **421** that indicates available PLMNs, network slices, and features of the PLMNs and slices. The RRC in CU **451** broadcasts the SIB data over DU **441** and RU **431**, and UE **401** receives the wireless broadcast of the SIB data. The RRC in CU **451** receives a service request for PLMN A and slice B from the RRC in UE **401**. The RRC in CU **451** transfers the service request for PLMN A and slice B to wireless network core **421** and receives service instructions with UE context. In response to the service instructions, UE **401** and the SDAP in CU **451** exchange user data over RU **431** and DU **441** using PLMN A and slice B. The SDAP in CU **451** and wireless network core **421** exchange user data using PLMN A and wireless network slice B using PLMN A and slice B to serve user application **405** in UE **401**.

(36) FIG. 7 illustrates a wireless network core **421** that serves 5G UE **401** which optimizes network communications for user application **405**. Wireless network cores **422-423** could be similar. Wireless network core **421** comprises Network Function Virtualization Infrastructure (NFVI) hardware **701**, NFVI hardware drivers **702**, NFVI operating systems **703**, NFVI virtual layer **704**, and NFVI Virtual Network Functions (VNFs) **705**. NFVI hardware **701** comprises Network

Interface Cards (NICs), CPU, RAM, Flash/Disk Drives (DRIVE), and Data Switches (SW). NFVI hardware drivers **702** comprise software that is resident in the NIC, CPU, RAM, DRIVE, and SW. NFVI operating systems **703** comprise kernels, modules, applications, containers, hypervisors, and the like. NFVI virtual layer **704** comprises vNIC, vCPU, vRAM, vDRIVE, and vSW. NFVI VNFs **705** comprise Access and Mobility Management Function (AMF) **771**, Uniform Data Management (UDM) **772**, Network Slice Selection Function (NSSF) **773**, Policy Control Function (PCF) **774**, Session Management Function (SMF) **775**, and User Plane Function (UPF) **776**. Other VNFs like Authentication Server Function (AUSF) and Network Repository Function (NRF) are typically present but are omitted for clarity. Wireless network core **421** may be located at a single site or be distributed across multiple geographic locations. The NIC in NFVI hardware **701** are coupled to RAN **411** and to external systems. NFVI hardware **701** executes NFVI hardware drivers **702**, NFVI operating systems **703**, NFVI virtual layer **704**, and NFVI VNFs **705** to serve UE **401** over RAN **411**.

(37) FIG. **8** further illustrates 5G wireless network core **421** that serves 5G UE **401** which optimizes network communications for user application **405**. AMF **771** performs SIB broadcasting, N2/N1 termination, N1 ciphering & integrity protection, UE registration. SMF/PCF selection, UE connection/mobility management. UE authentication and authorization, UE security management, and tracking area updates. User Data Management (UDM) **772** handles UE context, UE subscription data, and UE authentication keys. NSSF **773** performs slice discovery, AMF discovery, slice authorization, and slice prioritization. Policy Control Function (PCF) **774** distributes UE policies to the control plane based on network function. SMF **775** performs session establishment/management, network address allocation. N1 termination, downlink data notification, and traffic steering and routing. UPF **776** performs packet routing & forwarding, packet inspection, QoS handling, PDU interconnection, and mobility anchoring.

(38) AMF **771** receives attachment signaling for UE **401** from RAN **411** that requests PLMN A and slice B. AMF **771** interacts with UDM **772** to authenticate UE **401** and authorize PLMN A. AMF **771** interacts with NSSF **773** to authorize slice B. AMF **771**, PCF **774**, and SMF **775** interact to select the remaining UE context for the wireless data service for UE **401**. AMF **771** directs RAN **411** to serve UE **401** per the UE context which includes PLMN A and slice B. SMF **775** directs UPF **776** to serve UE **401** per the UE context which includes PLMN A and slice B. RAN **411** and UPF **776** serve UE **401** per the UE context by using PLMN A and slice B.

(39) FIG. **9** illustrates the operation of 5G UE **401** to optimize network communications for user application **405**. The operation is exemplary and may vary in other examples. In wireless UE **401**, operating system **404** directs a graphic display to prompt a user to obtain the user's express permission for user application **405** to access network characteristics for RANs **411-413**. Operating system **104** receives a user instruction through the display to permit user application **405** to access the network characteristics. The network characteristics comprise radio signal metrics, PLMN features, wireless network slice parameters, UE antenna configurations, RRC information, SIB data, and the like. For example, the RRC data may indicate received signal power and interference metrics for RANs **411-413**.

(40) To optimize network communications, user application **405** transfers a call to operating system **404** for the network characteristics. In response to the user permission and the call, operating system **404** transfers a request for the network characteristics to the RRC for radio **407**. The operation of the RRC for radios **408-409** is similar. The PHY in UE **401** detects the network characteristics from the downlink symbols provided by radio **407** and transfers the network characteristics to the RRC. The RRC responds to operating system **404** with the network characteristics. Operating system **404** transfers the network characteristics to user application **405**.

(41) User application **405** selects one or more of RANs **411-413**, wireless network slices, and/or PLMNs based on the network characteristics. In this example, user application **405** selects RAN **411**, PLMN A, and wireless network slice B. User application **405** transfers a call to operating

system **404** for RAN **411**, PLMN A, and slice B. Operating system **404** transfers a request for RAN **411**, PLMNs A, and slice B to the RRC that interacts with RAN **411**. The RRC in UE **401** exchanges signaling with the RRC in CU **451** to establish the wireless data service using PLMN A and slice B. The RRC in UE **401** exchanges signaling with the AMF **771** in core **421** to establish the wireless data service using PLMN A and slice B. AMF **771** interacts with UDM **772** to authenticate UE **401** and authorize PLMN A for UE **401**. AMF **771** interacts with NSSF **772** to authorize slice B for UE **401**. AMF **771**, PCF **774**, and SMF **775** interact to select the remaining UE context like QoS and network addresses for the wireless data service over PLMN A and slice B. AMF **771** directs the RRC in CU **451** to serve UE **401** per the UE context. SMF **775** directs UPF **776** to serve UE **401** per the UE context.

(42) In UE **401**, user application **405** exchanges user data with operating system **404**, and operating system **404** exchanges the user data with the SDAP for radio **407**. The SDAP for radio **407** and the SDAP in CU **451** exchange the user data per the UE context which requires PLMN A and slice B. The SDAP in CU **451** and UPF **776** exchange the user data per the UE context using PLMN A and slice B. UPF **776** exchanges the user data with external systems per the UE context using PLMN A and slice B.

(43) The wireless data network circuitry described above comprises computer hardware and software that form special-purpose UE circuitry to optimize network communications for user applications. The computer hardware comprises processing circuitry like CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory. To form these computer hardware structures, semiconductors like silicon or germanium are positively and negatively doped to form transistors. The doping comprises ions like boron or phosphorus that are embedded within the semiconductor material. The transistors and other electronic structures like capacitors and resistors are arranged and metallurgically connected within the semiconductor to form devices like logic circuitry and storage registers. The logic circuitry and storage registers are arranged to form larger structures like control units, logic units, and Random-Access Memory (RAM). In turn, the control units, logic units, and RAM are metallurgically connected to form CPUs, DSPs, GPUs, transceivers, bus circuitry, and memory.

(44) In the computer hardware, the control units drive data between the RAM and the logic units, and the logic units operate on the data. The control units also drive interactions with external memory like flash drives, disk drives, and the like. The computer hardware executes machine-level software to control and move data by driving machine-level inputs like voltages and currents to the control units, logic units, and RAM. The machine-level software is typically compiled from higher-level software programs. The higher-level software programs comprise operating systems, utilities, user applications, and the like. Both the higher-level software programs and their compiled machine-level software are stored in memory and retrieved for compilation and execution. On power-up, the computer hardware automatically executes physically-embedded machine-level software that drives the compilation and execution of the other computer software components which then assert control. Due to this automated execution, the presence of the higher-level software in memory physically changes the structure of the computer hardware machines into special-purpose UE circuitry to optimize network communications for user applications.

(45) The above description and associated figures teach the best mode of the invention. The following claims specify the scope of the invention. Note that some aspects of the best mode may not fall within the scope of the invention as specified by the claims. Those skilled in the art will appreciate that the features described above can be combined in various ways to form multiple variations of the invention. Thus, the invention is not limited to the specific embodiments described above, but only by the following claims and their equivalents.

Claims

1. A method to optimize network communications for a user application, the method comprising: generating network characteristics for multiple wireless communication networks based wireless signals received from the multiple wireless communication networks, wherein the network characteristics comprise one or more network slice parameters for network slices associated with the multiple wireless communication networks; transferring the network characteristics that include the one or more network slice parameters for the multiple wireless communication networks to the user application; receiving a selection of one of the multiple wireless communication networks from the user application based on the network characteristics that include the one or more network slice parameters; and communicating over a radio with the selected one of the multiple wireless communication networks responsive to the selection from the user application.
2. The method of claim 1 wherein the wireless signals received from the multiple wireless communication networks comprise System Information Blocks (SIBs).
3. The method of claim 1 wherein the wireless signals received from the multiple wireless communication networks comprise Public Land Mobile Network (PLMN) information.
4. The method of claim 1 wherein: receiving the selection of the one of the wireless communication networks from the user application comprises receiving an indication of a wireless network slice in the one of the wireless communication networks; and communicating over the radio with the selected one of the multiple wireless communication networks comprises communicating with the wireless network slice in the selected one of the multiple wireless communication networks.
5. The method of claim 1 wherein: receiving the selection of the one of the wireless communication networks from the user application comprises receiving an indication of a Public Mobile Land Network (PLMN) in the one of the wireless communication networks; and communicating over the radio with the selected one of the multiple wireless communication networks comprises communicating with the PLMN in the selected one of the multiple wireless communication networks.
6. The method of claim 1 wherein: receiving the selection of the one of the wireless communication networks from the user application comprises receiving an indication of a Radio Access Technology (RAT) in the one of the wireless communication networks; and communicating over the radio with the selected one of the multiple wireless communication networks comprises wirelessly communicating with the RAT in the selected one of the multiple wireless communication networks.
7. The method of claim 1 wherein: transferring the network characteristics for the multiple wireless communication networks to the user application comprises an operating system receiving an operating system call for the network characteristics from the user application and the operating system transferring the network characteristics for the multiple wireless communication networks to the user application in response to the operating system call; and receiving the selection of the one of the multiple wireless communication networks from the user application comprises the operating system receiving another operating system call from the user application to use the selected one of the multiple wireless communication networks.
8. One or more non-transitory machine-readable storage media that stores processing instructions that direct processing circuitry to perform a method to optimize network communications for a user application, the method comprising: generating network characteristics for multiple wireless communication networks based wireless signals received from the multiple wireless communication networks, wherein the network characteristics comprise one or more network slice parameters for network slices associated with the multiple wireless communication networks; transferring the network characteristics that include the one or more network slice parameters for the multiple wireless communication networks to the user application; receiving a selection of one of the multiple wireless communication networks from the user application based on the network characteristics that include the one or more network slice parameters; and communicating over a

radio with the selected one of the multiple wireless communication networks responsive to the selection from the user application.

9. The one or more non-transitory machine-readable storage media of claim 8 wherein the wireless signals received from the multiple wireless communication networks comprise System Information Blocks (SIBs).

10. The one or more non-transitory machine-readable storage media of claim 8 wherein the wireless signals received from the multiple wireless communication networks comprise Public Land Mobile Network (PLMN) information.

11. The one or more non-transitory machine-readable storage media of claim 8 wherein: receiving the selection of the one of the wireless communication networks from the user application comprises receiving an indication of a wireless network slice in the one of the wireless communication networks; and communicating over the radio with the selected one of the multiple wireless communication networks comprises communicating with the wireless network slice in the selected one of the multiple wireless communication networks.

12. The one or more non-transitory machine-readable storage media of claim 8 wherein: receiving the selection of the one of the wireless communication networks from the user application comprises receiving an indication of a Public Mobile Land Network (PLMN) in the one of the wireless communication networks; and communicating over the radio with the selected one of the multiple wireless communication networks comprises communicating with the PLMN in the selected one of the multiple wireless communication networks.

13. The one or more non-transitory machine-readable storage media of claim 8 wherein: receiving the selection of the one of the wireless communication networks from the user application comprises receiving an indication of a Radio Access Technology (RAT) in the one of the wireless communication networks; and communicating over the radio with the selected one of the multiple wireless communication networks comprises wirelessly communicating with the RAT in the selected one of the multiple wireless communication networks.

14. The one or more non-transitory machine-readable storage media of claim 8 wherein: transferring the network characteristics for the multiple wireless communication networks to the user application comprises an operating system receiving an operating system call for the network characteristics from the user application and the operating system transferring the network characteristics for the multiple wireless communication networks to the user application in response to the operating system call; and receiving the selection of the one of the multiple wireless communication networks from the user application comprises the operating system receiving another operating system call from the user application to use the selected one of the multiple wireless communication networks.

15. A wireless communication device to optimize network communications for a user application, the wireless communication device comprising: network circuitry configured to wirelessly receive wireless signals from multiple wireless communication networks, and in response, to generate network characteristics for the multiple wireless communication networks and transfer the network characteristics for the multiple wireless communication networks to processing circuitry, wherein the network characteristics comprise one or more network slice parameters for network slices associated with the multiple wireless communication networks; the processing circuitry configured to transfer the network characteristics that include the one or more network slice parameters to the user application and responsively receive a selection of one of the one of the multiple wireless communication networks from the user application based on the network characteristics that include the one or more network slice parameters; the processing circuitry further configured to exchange user data with the user application and with the network circuitry; and the network circuitry further configured to wirelessly exchange the user data with the selected one of the multiple wireless communication networks.

16. The wireless communication device of claim 15 wherein the wireless signals comprise System

Information Blocks (SIBs).

17. The wireless communication device of claim 15 wherein the wireless signals comprise Public Land Mobile Network (PLMN) information.

18. The wireless communication device of claim 15 wherein: the processing circuitry is configured to receive the selection of the one of the wireless communication networks comprises the processing circuitry configured to receive an indication of a wireless network slice in the one of the wireless communication networks; and the network circuitry is configured to wirelessly exchange the user data with the selected one of the multiple wireless communication networks comprises the network circuitry configured to wirelessly exchange the user data with the wireless network slice in the selected one of the multiple wireless communication networks.

19. The wireless communication device of claim 15 wherein: the processing circuitry is configured to receive the selection of the one of the wireless communication networks comprises the processing circuitry configured to receive an indication of a Public Mobile Land Network (PLMN) in the one of the wireless communication networks; and the network circuitry is configured to wirelessly exchange the user data with the selected one of the multiple wireless communication networks comprises the network circuitry configured to wirelessly exchange the user data with the PLMN in the selected one of the multiple wireless communication networks.

20. The wireless communication device of claim 15 wherein: the processing circuitry is configured to transfer the network characteristics for the multiple wireless communication networks to the user application comprises an operating system in the processing circuitry configured to receive an operating system call for the network characteristics from the user application and the operating system in the processing circuitry to transfer the network characteristics for the multiple wireless communication networks to the user application in response to the operating system call; and the processing circuitry is configured to receive the selection of the one of the multiple wireless communication networks from the user application comprises the operating system configured to receive another operating system call from the user application to use the selected one of the multiple wireless communication networks.
